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Calibrated Hallucination Risk for Large Language Models Under Domain and Task Shift with Evidence-Consistent XGBoost

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Abstract

Large language models answer fluently, but some answers are not supported by the evidence. Detectors based on a large judge model, an encoder, or model internals are expensive and rarely report calibration under domain or task shift. Feature-based detectors can be cheaper. Their calibration and sensitivity to answer length are usually not analyzed together. This work investigates whether a simple feature model can predict hallucination risk and how its ranking and calibration are affected by shift. This work introduces EC-XGB, an evidence-consistent XGBoost model with base features, claim-level natural language inference (NLI) aggregate scores, and a source indicator. The model is trained on HaluEval QA and RAGTruth non-QA rows with grouped splits and three seeds, and analyzed with fixed-hyperparameter ablations, source and target calibration, conformal prediction, Shapley additive explanations (SHAP) stability analysis, a length-only control model, and judge stress tests. A feature model is cheap and interpretable, and its inputs can be observed directly, making it a feasible choice for routine screening. On HaluEval QA, the fixed-hyperparameter model reaches F1 0.985 and AUROC 0.999, but lexical overlap and length carry that score. On MedHallu, it ranks the hallucinated answer higher than the grounded answer in 86.7% of 1,000 pairs, but flags every row and is poorly calibrated (ECE 0.493). On RAGTruth QA, the model stays close to chance. Target calibration lowers ECE from 0.731 to 0.125, but flags no target rows at the default threshold. Split conformal prediction reaches 0.944 coverage at 90%, with only 0.685 positive coverage. The dialogue dataset is almost completely flagged, summarization ranks lower than a length-only model, and a GPT judge beats the feature model on natural QA. Hence, EC-XGB is a domain-specific risk score, not a general-purpose detector.

1. Introduction

Black-box detectors evaluate candidate answers in their context [1, 2, 3, 4]. Detectors take the question, supporting evidence, and the candidate answer as input and output a score or a decision. Some detectors use a large judge model, some use an encoder on the input pair, and some analyze internal states. These detectors require heavy computation, and encoder-based methods are hard to interpret. Since the input is plain text, a detector can be placed before any model. Two recent reviews classify detectors [5, 6]. Evaluation and reliability of detectors remain open questions, and the third review identifies the reliability gap independently [7].

HaluAgent showed that small models could pick tools, though it required several calls per answer [8]. Valentin et al. designed a calibrated multi-score detector with a cost analysis, but it is based on LLM scoring [9]. Sundaragiri et al. proposed a retrieval, natural language inference (NLI), and XGBoost solution, but reported only classification metrics [10]. Haq et al. assigned SHAP-LIME explanations to words but did not measure their reliability [11]. In Deng et al.'s study of multi-turn dialogue, detector performance deteriorated outside the training domain [12]. FaithBench shows that current LLMs still produce summarization hallucinations, but it does not measure detector calibration [13]. Thus, four open problems remain in existing research. First, the calibration of lightweight detectors is rarely evaluated. Second, explanation quality is evaluated even less often. Third, few models have been evaluated on cross-domain behavior. Finally, error-pattern analysis on a per-example basis remains rare. The existing conventional feature-based benchmark also has several issues: it depends on answer length, does not give claim-level explanations, trains exclusively on HaluEval with a source-fitted calibrator, and lacks transferable probabilities [14]. HaluEval is saturated as well. There is still a gap in the form of a risk score that maintains rank and probability when the corpus changes. This paper fills these gaps with a feature-based model that scores a black-box LLM output. Given a query q, a reference

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context c, and a candidate answer a, it outputs $P(\text{hallucinated} \mid q, c, a)$. The EC-XGB model combines 26 features divided into seven groups, eight claim-level NLI aggregates, and a source indicator inside a tuned XGBoost classifier. The score is mapped to a probability with Platt calibration, and explanations are produced with Shapley additive explanations (SHAP). Grouped splits with length and overlap controls reduce shortcut learning. Source calibration is fitted on validation data only. Target-calibration experiments check the reliability of predictions. The log transform compresses the length scale but keeps the cue. The claim aggregates support per-claim explanations, but the seen-task gain on RAGTruth comes from the multi-source rows and source indicator, not the claim aggregates. Transfer limits are established on RAGTruth and FaithBench. The evaluation adds MedHallu [15], a HaluBench subset [16] with no training overlap, the HaluEval task shifts, conformal prediction, and judge stress tests. It reports F1, ranking, calibration, and paired errors, separating ranking from calibration: the model transfers ranking to medical pairs but not its probability, and it fails under the task shift.

HaluEval QA provides 10,000 question pairs, each with a correct and a hallucinated answer, for training and in-domain testing. RAGTruth provides natural responses for adaptation and recalibration. The target application is question answering with a reference context, and FaithBench serves only as an out-of-domain summarization stress test. The fixed-hyperparameter EC-XGB stack (m3) reaches F1 0.9854 and AUROC 0.9986 on HaluEval QA. RAGTruth shows the same pattern: the flag rate falls on tasks seen in training, while held-out QA stays near chance. On MedHallu, the same model ranks the hallucinated answer above the grounded one in 86.7% of 1,000 pairs but flags every row. On HaluEval dialogue it flags almost every row, and on HaluEval summarization it ranks below a length-only model. Target calibration on 5,034 RAGTruth QA calibration rows lowers ECE from 0.731 to 0.125 without changing the ranking. One answer takes 127.5 ms on GPU (2.00 s on CPU), and the approximate cost is about $0.001 per 1,000 predictions.

2. Conventional Method

2.1. Preliminaries

This section covers the traditional feature-based detector along with its evaluation metrics. The hallucinations are the answers that cannot be supported by the reference context or contradict it. Every data row is in the form of a tuple (q_i, c_i, a_i) comprising the question, reference context, and candidate answer with the label $y_i \in \{0, 1\}$. The traditional way of evaluating the detector is to use any one of the four metrics: precision, recall, F1, or AUROC of the labeled tuples. There are two additional requirements that must be satisfied by the deployed detector. Calibration demands that the risk score should match the observed frequency of the positive outcome, which is summarized by ECE [17]. Explanation means that the detector must point out the features that contributed to the warning, which is provided by SHAP for tabular models [18].

2.2. Baseline Method

The conventional method is a tuned XGBoost classifier [19] over the 26 base features in Table 1, trained on HaluEval QA. XGBoost models nonlinear interactions and gives exact TreeSHAP values, which the explanation analysis needs. A published feature-based detector uses the same learner with retrieval and NLI features [10], so this model is the natural baseline. Trees are added one at a time, and each tree fits the current gradient and curvature of the loss. A row enters at the root of each tree, follows the split conditions to one leaf per tree, and the leaf weights are summed into a raw margin. A logistic link maps the raw margin to a probability.

Let $x_i \in \mathbb{R}^{26}$ be the feature vector of row i, $y_i \in \{0, 1\}$ its label, and n the number of training rows. The model sums K trees into a raw margin, $z_i = \sum_{k=1}^{K} f_k(x_i)$. The training objective combines a differentiable loss with a regularizer over the trees,

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, z_i) + \sum_{k=1}^{K} \left(\gamma T_k + \tfrac{1}{2}\lambda \|w_k\|^2\right), \tag{1}$$

where $\phi = \{f_k\}_{k=1}^{K}$ collects the trees, f_k is the k-th tree with T_k leaves and leaf-weight vector $w_k \in \mathbb{R}^{T_k}$, γ is the penalty per leaf, and λ is an L2 weight penalty. The loss $l(y_i, z_i)$ is the logistic loss on the raw margin. The probability is the sigmoid of the raw margin, $\hat{p}_i = \sigma(z_i) = 1/(1 + e^{-z_i})$.

At boosting step t, a new tree f_t with T leaves and leaf weights $w_1, \dots, w_T$ is added, and the objective is approximated, up to a constant, by a second-order expansion around the current raw margin,

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[g_i f_t(x_i) + \tfrac{1}{2} h_i f_t^2(x_i) \right] + \gamma T + \tfrac{1}{2} \lambda \sum_{j=1}^{T} w_j^2, \tag{2}$$

where $g_i = \partial l(y_i, z)/\partial z$ and $h_i = \partial^2 l(y_i, z)/\partial z^2$ are evaluated at $z_i^{(t-1)}$, and $f_t(x_i) = w_{q(x_i)}$ is the leaf weight that row i reaches, with $q(x_i)$ the leaf index. For the logistic loss, $g_i = \hat{p}_i - y_i$ and $h_i = \hat{p}_i(1 - \hat{p}_i)$.

For a fixed tree structure, the optimal weight of leaf j is

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \tag{3}$$

where $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$ are the sums over the rows I_j that reach leaf j. The gain of a split is

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma, \tag{4}$$

with G_L and H_L for the left child and G_R and H_R for the right child. A split is kept only when the gain is positive. The tree also stops at the maximum depth or when a leaf has fewer than the minimum child weight.

Depth, learning rate, tree count, and the row and column subsampling rates are tuned on the 14,000 training rows. Each seed runs its own 30-configuration search with stratified group five-fold cross-validation, and the grouping keeps both answers to the same question in one fold. The seed-42 values are in Section 4, and the per-seed values are in the repository. Each seed trains on the 14,000 rows with early stopping on the 3,000 validation rows (30 rounds). The 300-tree setting is the tuned upper bound, and seed 42 stops at 209 trees. The headline comparison uses the uncalibrated probability at a fixed 0.5 threshold on the 3,000 test rows. Platt and isotonic maps are fitted on the same validation rows for the calibration study. The source-only P0 threshold is selected on those rows at a 5% false-positive budget. The test rows are never used for tuning, early stopping, calibration, or threshold selection. The final step returns the mean over the three seeds. The search leaves γ, λ, and the minimum child weight at their XGBoost defaults. Algorithm 1 summarizes the tuning, training, and calibration steps.

Algorithm 1 Baseline feature-based XGBoost (training and evaluation)

Require: labeled rows (q_i, c_i, a_i, y_i) and seeds $S = \{42, 123, 456\}$
Ensure: trained model, test metrics, and Platt and isotonic maps

```
extract the 26 base features for every row
for each seed s ∈ S do
    tune XGBoost on the 14,000 training rows with 30 random configurations and stratified group 5-fold cross-validation
    train the classifier on the same rows with early stopping on the 3,000 validation rows
    evaluate the raw score on the 3,000 test rows at the fixed 0.5 threshold
    fit Platt and isotonic maps on the validation rows for the calibration study
end for
return the mean metrics over the seeds
```

3. Proposed Method

3.1. Working Principle

The aim is to predict $P(\text{hallucinated} \mid q, c, a)$, where q is the input question, c is the optional context, and a is the answer generated by the LLM. Truthfulness is not the concern here, but rather the assessment of evidence consistency, that is, whether a is entailed by c or contradicts c. The output of the system is a probability value, which is translated to a label through the decision threshold. Figure 1 shows the pipeline used to calculate the probability. The pipeline takes the inputs and extracts 35 features: 26 base features split into seven groups, eight aggregate NLI scores at the

claim level, and a source indicator. EC-XGB takes the features and generates a raw score, which is calibrated by a Platt calibrator trained on RAGTruth QA calibration rows to obtain the displayed probability. The model outputs a risk score, verdicts at the claim level, and SHAP values. The bottom part of the figure shows the deployment pipeline, where the Next.js front end calls the FastAPI service, which loads the model, calibrator, explainer, and feature models at startup. The right block lists the evaluation datasets. HaluEval evaluates in-domain performance, while RAGTruth summarization and data-to-text measure performance on task types seen in training. RAGTruth QA and FaithBench are held out.

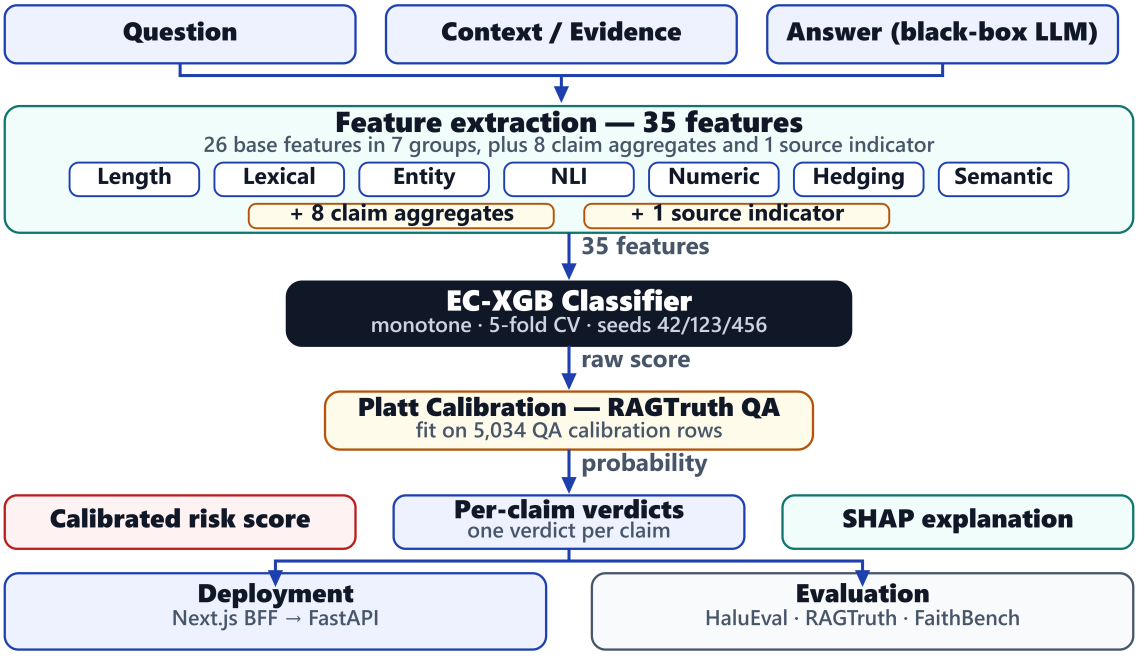


Figure 1: End-to-end pipeline, deployment path, and evaluation corpora of EC-XGB.

3.2. Features

The model uses 26 base features in seven groups (Table 1) and eight claim-level aggregates (Table 2). NLI probabilities come from the DeBERTa-v3 cross-encoder [20] trained on SNLI [21] and MultiNLI [22]. Semantic similarity is measured with MiniLM embeddings [23, 24], and entities are found through spaCy named entity recognition (NER) [25]. Length is used since HaluEval labels longer answers as hallucinated more often, making length a powerful but fragile feature. Lexical overlap indicates copying and is the strongest SHAP feature; nevertheless, it does not detect a fluent paraphrase that contradicts the evidence. A novel entity is a useful warning sign, although a new name can be part of a correct answer. The NLI features are the only group that provides logical support and not just overlap. A changed number is a common and easy-to-detect case of hallucination. Although hedging words are weak on their own, they make it possible to tell apart a cautious correct answer from a confident wrong one. Semantic similarity finds paraphrase and topic changes missed by lexical overlap. The eight claim-level aggregates repeat NLI features at the level of atomic claims, which is the level reviewers usually operate at. RAGChecker uses entailment-based claim metrics [26].

3.3. Evidence-Consistent XGBoost

EC-XGB modifies the conventional setup in terms of features and training samples. The character and word counts are log-transformed, and monotone constraints are placed on the evidence features. The constraints are negative for context overlap, context Jaccard, context entailment, context cosine, claim minimum and mean entailment, and the supported-claim ratio, and positive for context contradiction, novel-entity ratio, novel numbers, claim maximum and mean contradiction, and the contradicted-claim ratio. All other features remain unconstrained. Eight claim features

Table 1
Base feature groups (26 features total).

Group	Features	Source
Length/style	n_chars, n_words, n_sentences, avg_word_len	regex
Lexical	overlap vs. context & question, Jaccard ×2	token sets
Entity	entity counts, overlap ratio, novel-entity ratio	spaCy NER
NLI	entail/contradict/neutral × 2 directions	cross-encoder NLI
Numeric	number counts, overlap ratio, novel numbers	regex
Hedging	hedge count, hedge density	lexicon
Semantic	cosine(context,answer), cosine(question,answer)	MiniLM

Table 2
Claim-level NLI aggregates (8 features).

Signal	Features	Source
Contradiction	max and mean contradiction, contradicted-claim ratio	NLI cross-encoder
Support	supported-claim ratio, unsupported-claim ratio, min and mean entailment	NLI cross-encoder
Decomposition	number of claims	sentence splitter

from Table 2 are included: seven NLI aggregates and the claim count. They use the same NLI scorer and the support-first rule as the verification layer. A claim is supported if there is a retained sentence with an entailment probability of at least 0.5 that is at least as large as its contradiction probability. It is contradicted if there is no supporting sentence and there is a sentence with a contradiction probability of at least 0.5 that is larger than its entailment probability. Otherwise, the claim is considered unsupported.

Consider an answer with K claims. Let $C(k)$ be the maximum contradiction probability of claim k over its selected context sentences, $E(k)$ the maximum entailment probability, and v_k its verdict. The aggregate vector is

$$\psi(x) = [\max_k C(k), \overline{C}, r_{\text{con}}, r_{\text{sup}}, r_{\text{unsup}}, \min_k E(k), \overline{E}, K], \tag{5}$$

with

$$r_{\text{con}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{contradicted}], \quad r_{\text{sup}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{supported}], \quad r_{\text{unsup}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{unsupported}]. \tag{6}$$

Here $x = (q, c, a)$ is the input triplet, and $\overline{C}$ and $\overline{E}$ are the means of $C(k)$ and $E(k)$ over the claims. For $K > 0$, the three ratios sum to one. The model input has 35 dimensions: the 26 base features (two of them log-scaled), the eight claim features, and one source indicator.

The training sample is extended to include the RAGTruth summarization and data-to-text rows. The ablation starts with m0, the 26-feature conventional reference, and moves on to m1, which uses log-transformed length counts under monotone constraints, to m2, which adds the eight claim features (34 features total), and to m3, which uses the RAGTruth non-QA rows with the source indicator (35 features total). A binary source indicator equal to one is used for those rows and equal to zero for HaluEval, which allows the classifier to separate the two sources. The classifier does not see RAGTruth QA rows and thus performs zero-shot QA ranking. In contrast, the display calibrator sees labeled QA rows and produces a non-zero-shot display score. The 35-feature setup is called EC-XGB, with seed 42 as

the deployed checkpoint and other seeds as stability checks. The ablation described in Section 5 shows that the claim block makes no aggregate F1 contribution, and the claim count K is kept because answers with more claims are harder to verify. Algorithm 2 describes the procedure: steps 1 and 2 generate the feature matrix, and steps 3 to 9 perform the model fitting for the three seeds. Each iteration generates the 35-feature input, trains on HaluEval plus the RAGTruth non-QA rows, fits the display calibrator on the 5,034 QA calibration rows, and sets the source and target thresholds.

Algorithm 2 Evidence-consistent XGBoost (EC-XGB) training and deployment

Require: HaluEval QA training and validation rows, RAGTruth summarization and data-to-text training rows, RAGTruth QA calibration rows, seeds $\mathcal{S} = \{42, 123, 456\}$
Ensure: seed-42 model and display calibrator, source and target thresholds, per-seed metrics
1: extract the 26 base features and the eight claim features for every row
2: log-scale the character and word counts, and attach the monotone constraints
3: **for** each seed $s \in \mathcal{S}$ **do**
4: build the 35-feature input, with the source indicator at zero for HaluEval and one for RAGTruth
5: train XGBoost on the HaluEval rows plus the RAGTruth non-QA rows, and keep every RAGTruth QA row out of classifier training
6: fit the Platt display calibrator on the RAGTruth QA calibration rows
7: select the source threshold on the HaluEval validation split
8: select the target threshold on the QA calibration rows at a 5% false-positive budget
9: **end for**
10: **return** the seed-42 model and calibrator, both thresholds, and the per-seed metrics

3.4. Calibration and Explainability

The source calibration uses the HaluEval validation split. Platt scaling fits a sigmoid mapping $\hat{p} = \sigma(as + b)$ to the uncalibrated probability s, where a and b are learned by maximum likelihood [27]. Isotonic regression learns a non-decreasing step function g such that $\hat{p} = g(s)$ [28]. Both methods are tested against the uncalibrated probability on the HaluEval test split. The comparison provides the expected calibration error measured over 10 equal-width bins,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{7}$$

where $\mathrm{acc}(B_m)$ and $\mathrm{conf}(B_m)$ are the empirical positive rate and mean confidence in bin B_m, the Brier score $\mathrm{BS} = \frac{1}{n}\sum_i (\hat{p}_i - y_i)^2$, and reliability diagrams [17]. Post-hoc calibration degrades when the target distribution moves [29].

In the second protocol, the calibrator is trained on labeled target rows and tested on a disjoint target test split. The deployed EC-XGB display score is calibrated using a Platt calibrator fitted on the 5,034 RAGTruth QA calibration rows. These rows have the source indicator set to one, and the API uses the same value for all requests. This target calibration requires labeled target rows and is not zero-shot. The source-only protocol (P0 in Table 4) fits the calibrator and selects the threshold on the HaluEval validation split. The deployed calibrator is Platt. Platt versus isotonic was compared on the test split in an earlier deployment; see Section 5.

TreeExplainer computes the exact SHAP values [18] of the raw XGBoost model. The global importance of a feature is defined as the mean absolute SHAP value over a stratified test sample. The explanation endpoint returns five features with the largest absolute SHAP values for a given prediction. Post-hoc attribution scores can be unstable even when the underlying model is stable [30], so the pilot evaluates them directly [31]. A 20-case pilot evaluates the effect of six operators: entity, number, and date replacements, support removal, context insertion, and clause shuffling. It measures the mean absolute score change, the fraction of large changes, the top-1 flip rate, and the Spearman rank correlation between the original and perturbed rankings.

3.5. System and Reproducibility

The method is used in a web application with a Next.js frontend and a FastAPI backend. The browser calls the Next.js backend-for-frontend (BFF) which streams chat tokens and forwards tool calls to the model. The analyze route calls FastAPI via the `/api/ml/*` rewrite. At startup, FastAPI loads the EC-XGB model, the Platt calibrator, the SHAP explainer, and the NLI, spaCy, and MiniLM feature models. The deployed checkpoint corresponds to the 35-feature m3

model with seed 42. Section 5 tests this model against B2, the conventional 26-feature checkpoint trained on HaluEval QA, and against B6 m0, the fixed-hyperparameter 26-feature reference from the same tuning protocol. Claims are separated from the answer and verified with sentence-level NLI against the context, a local passage index, or a web search API. A claim is supported if at least one retained sentence provides an entailment probability of at least 0.5 and not lower than its contradiction probability. A contradiction decides only if no sentence supports the claim and a sentence has a contradiction probability of at least 0.5 that is higher than its entailment probability. A relevance gate filters out sentences that do not contain claim vocabulary, and verdicts are returned for each claim. The card shows these verdicts as follows: a contradicted claim is considered high-risk, an unsupported claim is medium-risk, and a fully supported answer is low-risk. The percentage is the display score. The chat card adjusts this score based on the claim mixture, and the analyze card reports the unadjusted Platt score. Figure 1 shows the scoring procedure for a single answer and Figure 2 shows where each component runs.

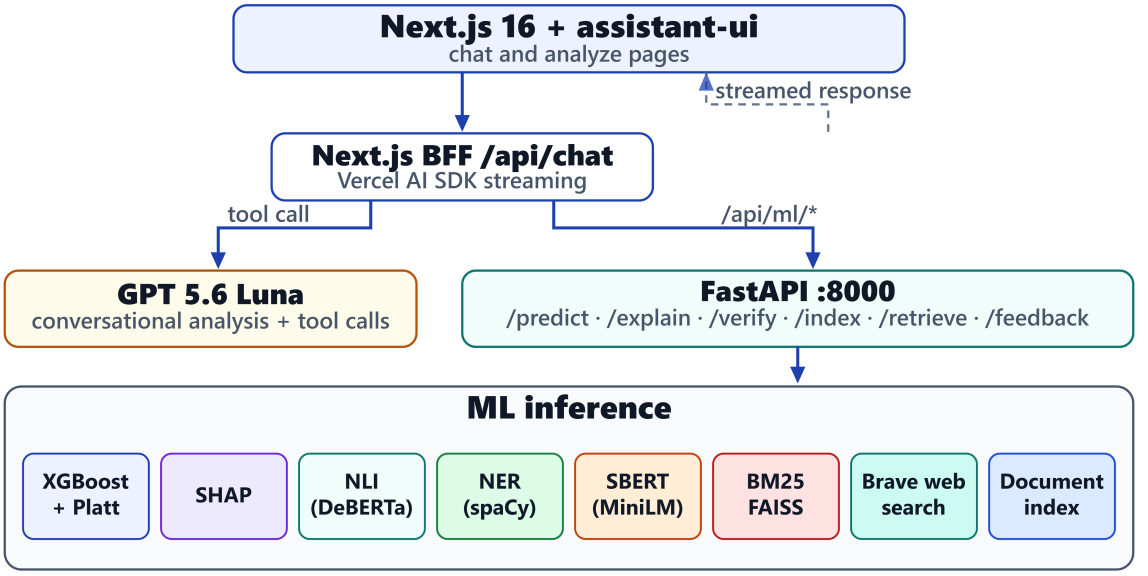


Figure 2: Deployment architecture of the HaluRISC system.

In the example of the hallucinated number, the context says 5 days per decade whereas the answer says 10. The raw score is 0.999 and the target-calibrated score is 35.4%, which the card rounds to 35% and puts in the medium band (0.35 to 0.60). The claim view marks the number claim unsupported, hence the chat card reports medium risk. Although the two labels agree in this case, they come from different rules. The claim verdict audit is reported in Section 5. The requirements file fixes all dependencies; split indices are saved with the artifacts; and the audit scripts check that rebuilt and reported values match the frozen artifacts. Audits run after each build and fail if there is any difference. The contribution is in the training and evaluation protocol, not in the development of a new learner. EC-XGB uses three standard changes: log-scaled length counts with monotone constraints, claim-level NLI aggregations for explanation, and multi-source training with a source indicator. Section 5 measures what each change contributes. The claim aggregations are kept for explaining each claim and do not contribute to any additional F1 gain. The seen-task gain comes from the m3 stack that adds the RAGTruth rows and the source indicator, and the two are not disentangled. No call to the LLM judge is required for the risk score. The claim judge provides an additional verification step, but the reported EC-XGB results do not depend on it.

4. Experiments

4.1. Experimental Setup

All experiments were done on a Windows laptop equipped with an NVIDIA RTX 3060 6 GB GPU, CUDA 12.8, torch 2.11.0+cu128, and 32 GB of RAM. The code is executed using Python 3.12 with scikit-learn 1.9.0, XGBoost 3.3.0, SHAP 0.52.0, spaCy 3.8.14, and sentence-transformers 5.6.1. Each experimental run can be reproduced owing

to stored split indices and centrally set seeds. A random search was done across 30 hyperparameter configurations using grouped five-fold cross-validation. For the seed-42 conventional model, the hyperparameter configuration has a maximum tree depth of 5, a learning rate of 0.05, 300 trees, a subsampling ratio of 0.7, and a column sampling ratio of 0.7. Each seed was independently tuned for the main experiment, and the ablation in Section 5 holds one configuration fixed in all variants so that feature effects are isolated from tuning. Seeds 42, 123, and 456 are repeated where a seed applies, and the single-seed tables are labeled.

4.2. Dataset

HaluEval QA [32] consists of 10,000 question-answer pairs, with a knowledge passage, a correct answer, and a hallucinated answer provided. Data is divided with a 70/15/15 split by item so that both answers associated with the same question are kept together. HaluEval is entirely generated by ChatGPT [32, 33], making the difference between factual inaccuracies and hallucinations somewhat blurry. RAGTruth [34] provides natural answers with human span annotations, and FaithBench [13] provides hallucinations for contemporary LLM summaries. RAGTruth maintains its official document-level partition: summarization and data-to-text provide training rows and a held-out test set each, and QA provides 5,034 calibration rows and a held-out test set. RAGTruth QA is a subset of the full corpus, so its rows are not additive. Three held-out test sets make up the pooled RAGTruth test set. Table 3 lists the corpora, their sizes, and positive rates. MedHallu pqa_labeled [15] adds medical QA pairs with human-annotated gold answers and paired LLM hallucinations. HaluBench [16] provides a stratified 30% sample of its DROP, FinanceBench, and covidQA rows, balanced between PASS and FAIL; its HaluEval, RAGTruth, and PubMedQA rows are excluded. Seeded samples of HaluEval tasks are used as task-shift sets. External corpora are never used for fitting or threshold selection.

Table 3
Datasets used for training and evaluation.

Corpus	Rows	Groups	Positive	Use
HaluEval QA	20,000	10,000	50.0%	train / validation / test
RAGTruth QA	5,934	989	17.8 / 31.1%	held-out test + calibration
RAGTruth full	17,790	2,965	34.9%	train + pooled test (2,700)
FaithBench	750	750	68.1%	zero-shot
MedHallu	2,000	1,000 pairs	50.0%	zero-shot external
HaluBench subset	900	3 sources	50.0%	zero-shot external
HaluEval dialogue	2,000	1,000 pairs	50.0%	task-shift test
HaluEval summarization	2,000	1,000 pairs	50.0%	task-shift test

Leading and trailing whitespace is trimmed, and lexical matching is case-insensitive. NER and NLI are performed on the original text, with entity strings converted to lowercase for matching. Missing context is a special case where lexical overlap is zero and NLI probabilities are uniformly distributed. In m1 and later variants, both length counts are log1p-transformed; in m0 and the conventional model, raw counts are used. Claims are split into no more than six clause-level segments, with conjunctions split into clauses. If the context has more than four sentences, each claim is matched to its four best sentences based on token overlap, and both sides of a pair are truncated to 400 characters. Both index sets are frozen. A RAGTruth answer is considered positive if it includes at least one annotated hallucination span. Severity annotations in FaithBench are condensed by a worst-severity principle, and a Questionable annotation is counted as positive. HaluEval uses its own label scheme. NLI and MiniLM inference ran on the reference GPU in fp16, and lexical and entity features were calculated on the CPU. The feature-based logistic baselines standardize their inputs on the training split; TF-IDF models use sparse features, and XGBoost does not scale. The source indicator takes a value of 1 for FaithBench and MedHallu, and 0 for HaluBench and the task-shift sets. FaithBench, RAGTruth summarization, and data-to-text have no question; therefore, question overlap is 0, and question similarity is defined as the cosine similarity between the answer and the empty string encoding. Licenses and the data availability statement are in the Declarations.

4.3. Baselines and Experimental Procedure

Twelve models are used in the evaluation. The heuristic assigns risk as one minus lexical overlap, and the length-only logistic control uses two log-scaled length counts. The classical models include logistic regression and a 300-tree random forest. The conventional XGBoost model [19] uses all 26 base features. Three more controls investigate the

feature design: NLI-only, answer-only TF-IDF, and full-input TF-IDF. The set is completed by the majority classifier, fine-tuned RoBERTa-base [35], and the light reimplementation of the ensemble of Sundaragiri et al. [10]; the two judges are presented in Section 5. Internal-state detectors are beyond the scope of the black-box approach [36]. The RoBERTa baseline processes up to 256 tokens and trains for three epochs. The majority baseline predicts the negative class, and, therefore, probability metrics are not applicable. The external ensemble uses five features (ROUGE-L, MiniLM cosine similarity, two NLI probabilities, answer length) in the logistic regression model trained on 14,000 training samples; the selected threshold of 0.55 gives F1 = 0.9514, so the fixed 0.5 threshold does not affect the comparison. EC-XGB's display calibrator is fit on the RAGTruth QA split, and the in-domain Platt calibrator is fit on the validation set. Metrics are rounded to four decimals and rates to one. Significance uses McNemar's test on the seed-42 predictions, 2,000 two-level bootstrap resamples, and Wilcoxon tests across seeds. No question appears in both training and evaluation. The calibration quality is estimated on raw, Platt, and isotonic scores; see Table 4 for definitions of P0, P1, and P2; the test rows are the same for all protocols.

Table 4
The three evaluation protocols.

Protocol	Calibrator fit	Threshold fit	Zero-shot
P0 source-only	HaluEval validation (3,000)	HaluEval validation	yes
P1 target-calibrated	RAGTruth QA calibration (5,034)	fixed at 0.5	no
P2 target-thresholded	RAGTruth QA calibration (5,034)	same rows at 5% FPR	no

The deployed system uses P1, and P2 is its thresholded version. Only P0 is a zero-shot model; P1 and P2 read labels from the target calibration split, which is disjoint from the test split. Global and local explanations are provided by the SHAP [18] TreeExplainers. Reliability is tested with two checks: rank agreement of the SHAP importances of the m3 and B2 models and a perturbation pilot using six operators. The feature latency is estimated on 200 HaluEval samples on the reference laptop. The 900 RAGTruth QA rows are scored by two judges, GPT-5.6 Luna and Qwen 3.5 4B, according to the LLM-as-a-judge protocol [37]; the costs are derived from the token counts.

4.4. Evaluation Metrics

Two types of metrics are introduced here. The first set includes the threshold-dependent metrics for evaluating the decision made at the fixed cut-off point. The second set contains the threshold-free metrics for evaluating how well the model ranks responses. Both are needed since the model can have high ranking performance and poor decision performance if the cut-off point is fixed. Precision is the proportion of flagged responses that were hallucinated. Recall is the proportion of hallucinated responses detected by the system, and F1 is the harmonic mean of precision and recall. False-positive rate (FPR) is the proportion of correct responses that get flagged and serves as an operating constraint under distributional shift. True positive (TP) and true negative (TN) indicate the correctly classified instances, and false positives (FP) and false negatives (FN) are a correct answer flagged as hallucinated and a hallucinated answer that is not flagged, respectively. The flag rate is the proportion of responses exceeding the decision threshold; if it approaches 100%, then every response reaches a human reviewer.

$$\text{Precision } (P) = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall } (R) = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad F_1 = \frac{2PR}{P + R}, \tag{8}$$

$$\text{False-positive rate (FPR)} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \qquad \text{Flag rate} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\hat{p}_i \geq \tau]. \tag{9}$$

The second set includes ranking, calibration, and paired error metrics. The area under the ROC curve (AUROC) is equal to the probability that a random hallucinated answer has a higher score than a random correct answer, with 0.5 meaning chance level. Area under the precision-recall curve (PR-AUC) is more informative when the positive class is rare, and RAGTruth QA has a positive rate of 17.8%. Thus, PR-AUC is also reported. The Matthews correlation coefficient (MCC) uses all four cells of the confusion matrix and has values ranging from −1 to 1, with zero being the value of random performance. The Wilson interval gives a 95% confidence range for a proportion and is used for the P2 recall and FPR when the number of observations is small.

$$\text{AUROC} = \int_0^1 \text{TPR}(u)\, d\, \text{FPR}(u) = P(s^+ > s^-), \tag{10}$$

$$\text{PR-AUC} = \int_0^1 P(r)\,dr, \qquad \text{MCC} = \frac{\text{TP}\cdot\text{TN} - \text{FP}\cdot\text{FN}}{\sqrt{(\text{TP}+\text{FP})(\text{TP}+\text{FN})(\text{TN}+\text{FP})(\text{TN}+\text{FN})}}, \tag{11}$$

$$\text{CI}_{95\%} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}. \tag{12}$$

5. Results

5.1. Model Comparison

Table 5 shows the HaluEval evaluation results for eleven baselines and EC-XGB averaged over three seeds where relevant. EC-XGB outperforms the conventional model in F1 by 0.0009, an improvement smaller than the width of the conventional seed-42 bootstrap confidence interval [0.9812, 0.9895], such that the benchmark cannot provide a definite order. The ranking columns show a comparable trend: EC-XGB takes first place among the feature-based models in AUROC and PR-AUC, with the conventional model and random forest close behind, and RoBERTa slightly above. Calibration paints a different picture, as the conventional model takes the lowest expected calibration error (ECE) and EC-XGB and RoBERTa take the lowest Brier scores. The weak controls are NLI-only and full-input TF-IDF, whereas answer-only TF-IDF is much stronger; adding question and context lowers performance. The reduced external ensemble is behind EC-XGB in F1 by 0.0337 [0.0262, 0.0419] and 0.0135 AUROC [0.0098, 0.0177], but it is trained on fewer rows and with a narrower feature set, so this gap is descriptive. Figure 3 shows the curves that underlie the table; the curves correspond to seed 42, while the table values correspond to three-seed averages.

The XGBoost margin relative to the random forest is 0.0004 F1, McNemar $p = 0.502$, meaning that there is no statistically significant difference. The margin over logistic regression is 0.0187 F1 ($p = 1.86 \times 10^{-10}$), and over the heuristic is 0.0495 ($p = 2.28 \times 10^{-30}$). A source-only encoder wins in-domain on F1 (0.9912 vs 0.9855) but is near chance on each RAGTruth task (AUROC 0.458 on summarization and 0.491 on data-to-text, compared to 0.736 and 0.770 for EC-XGB on those tasks). Pooling gives 0.337 for the encoder and 0.582 for EC-XGB, as the encoder's pooled score is a blend of between-task score shifts and within-task ranking. An adapted encoder on the same multi-source rows reaches 0.694 on the pooled RAGTruth test set, beating EC-XGB's 0.582; hence, the seen-task gain is not specific to the feature model. Neither model generalizes to unseen QA (0.490 vs 0.549), and FaithBench is worse for both (0.434 vs 0.594). Adaptation helps any model on seen tasks; the feature model preserves its relative ordering on unseen tasks.

Table 5
Test-set performance on HaluEval QA, mean over seeds 42/123/456 where a seed applies; deterministic rows are single fits (n = 3,000). †Reduced reimplementation of Sundaragiri et al.

Model	P	R	F1	AUROC	PR-AUC	MCC	ECE	Brier
Heuristic (1−overlap)	0.9473	0.9233	0.9352	0.9132	0.8196	0.8723	0.3541	0.2672
Length-only (2 feats)	0.9454	0.9460	0.9457	0.9759	0.9722	0.8913	—	—
Logistic regression	0.9768	0.9553	0.9660	0.9932	0.9901	0.9329	0.0099	0.0253
Random forest	0.9897	0.9789	0.9842	0.9980	0.9983	0.9687	0.0143	0.0127
NLI-only	0.6479	0.6967	0.6714	0.7177	0.7531	0.3189	0.0678	0.2182
TF-IDF (answer-only)	0.9550	0.8920	0.9224	0.9696	0.9665	0.8519	0.0831	0.0662
TF-IDF (full input)	0.6397	0.5647	0.5999	0.6754	0.6785	0.2484	0.0679	0.2328
External ensemble†	0.9508	0.9527	0.9517	0.9850	0.9856	0.9033	0.0141	0.0391
XGBoost (conventional)	0.9932	0.9762	0.9846	0.9979	0.9983	0.9697	0.0045	0.0117
EC-XGB (m3)	0.9939	0.9773	0.9855	0.9984	0.9987	0.9715	0.0074	0.0108
RoBERTa-base	0.9955	0.9869	0.9912	0.9990	0.9993	0.9825	0.0070	0.0081
Majority (negative)	0.0000	0.0000	0.0000	—	—	0.0000	—	—

5.2. EC-XGB and Ablation

Table 6 reports the ablation results for the model under the fixed hyperparameter configuration (max depth 5, learning rate 0.05, 300 trees, subsample 0.7, colsample 0.7) selected once in the m3 seed-42 run. The ladder from m0 to m3 is defined in Section 3, and ECE is computed on the raw score. The in-domain F1 span across m0 to m3

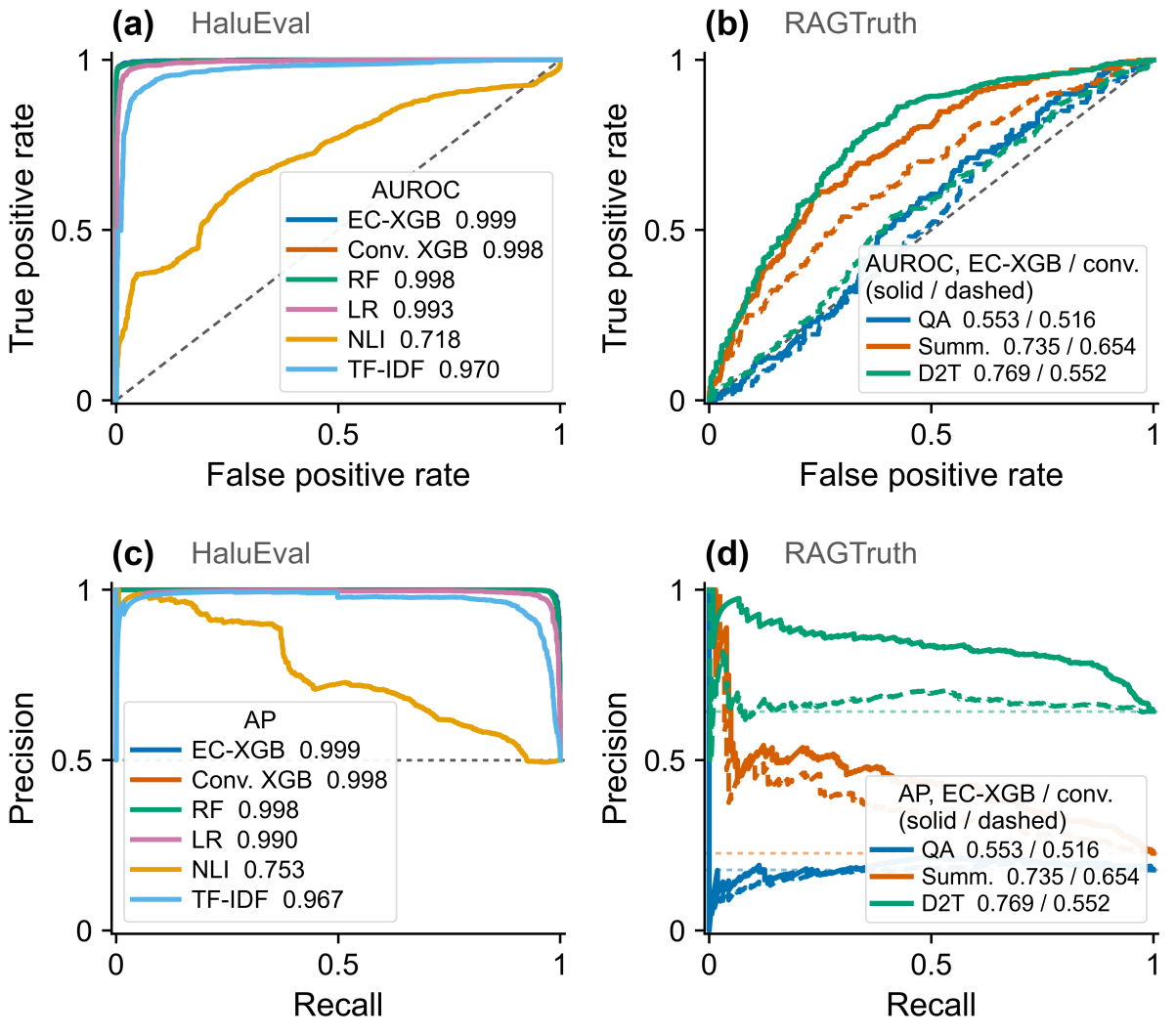


Figure 3: ROC and precision-recall curves on HaluEval and RAGTruth (seed 42).

is 0.0006. The claim aggregation changes F1 by −0.0004 at m2 compared to m1, and the entire m3 stack changes F1 by +0.0001; neither is an aggregate gain. On the pooled RAGTruth split, AUROC improves from 0.494 to 0.590 and raw ECE reduces from 0.635 to 0.275; for the held-out QA split the comparison is 0.556 vs 0.558, again within seed noise. McNemar yields $p = 0.789$ for m0 vs m3, so the ablation does not resolve a difference. The gain does not come from the claim features, but appears when the multi-source rows and source indicator are added. The MedHallu source-zero check suggests that the indicator is not the main driver of the gain, but a seen-task source ablation would settle it. The deployed checkpoint achieves 0.9855 with its own tuned hyperparameter configuration, 0.0001 above the fixed ablation.

Table 7 reproduces the ablation for the model under the fixed-hp configuration, whose full reference reaches F1 0.9861, AUROC 0.9985, and MCC 0.9726. In this setting, lexical overlap contributes the most, followed by length and NLI features. The claim aggregation block costs about one seed standard deviation, so an aggregate claim gain is not resolved. The entity, numeric, hedging, semantic, and source features do not change F1 by more than 0.0011. Thus, only lexical overlap and length carry the in-domain score, and the rest of the groups contribute ranking information without altering F1 substantially. Deltas are computed against the full m3 refit under the same configuration, trained

Table 6
EC-XGB component ablation on the HaluEval test set with one fixed configuration, mean over three seeds.

Variant	Features	F1	AUROC	MCC	ECE	RAGTruth AUROC	RAGTruth ECE
m0 conventional	26	0.9849	0.9979	0.9701	0.0050	0.4937	0.6352
m1 debias	26	0.9853	0.9976	0.9710	0.0042	0.5567	0.6260
m2 + claims	34	0.9849	0.9985	0.9701	0.0051	0.5461	0.6359
EC-XGB (m3)	35	0.9854	0.9986	0.9713	0.0078	0.5898	0.2746

Table 7
Feature-group ablation of the fixed-hp m3 model on the HaluEval test set, mean over seeds 42/123/456.

Removed group	Features	ΔF1	ΔAUROC	ΔMCC
length	4	−0.0066	−0.0012	−0.0127
lexical	4	−0.0171	−0.0040	−0.0343
entity	4	−0.0010	+0.0000	−0.0020
nli	6	−0.0040	−0.0004	−0.0080
numeric	4	−0.0003	+0.0001	−0.0007
hedging	2	+0.0000	+0.0001	0.0000
semantic	2	0.0000	+0.0002	+0.0000
claims	8	−0.0010	−0.0003	−0.0020
source	1	−0.0001	+0.0001	−0.0002

on the same 24,056 rows and evaluated once on 3,000 test rows without tuning or early stopping. The refit reference differs from Table 6 by 0.0007 F1, within a seed standard deviation of 0.0010.

Figure 4 shows the seed-level spread behind Table 6. The F1 and AUROC panels are tightly clustered, with within-variant differences below 0.0007 F1 and 0.0005 AUROC. The ECE panel is noisier: m0 is stable across seeds, while m2 and m3 vary more. The mean ordering is m3 > m1 > m2 > m0, but the per-seed ranks flip and the differences are below seed noise. The ECE sign is stable: m0 is lower than m3 on every seed, so the ablation does not establish a calibration advantage for m3 in this sample.

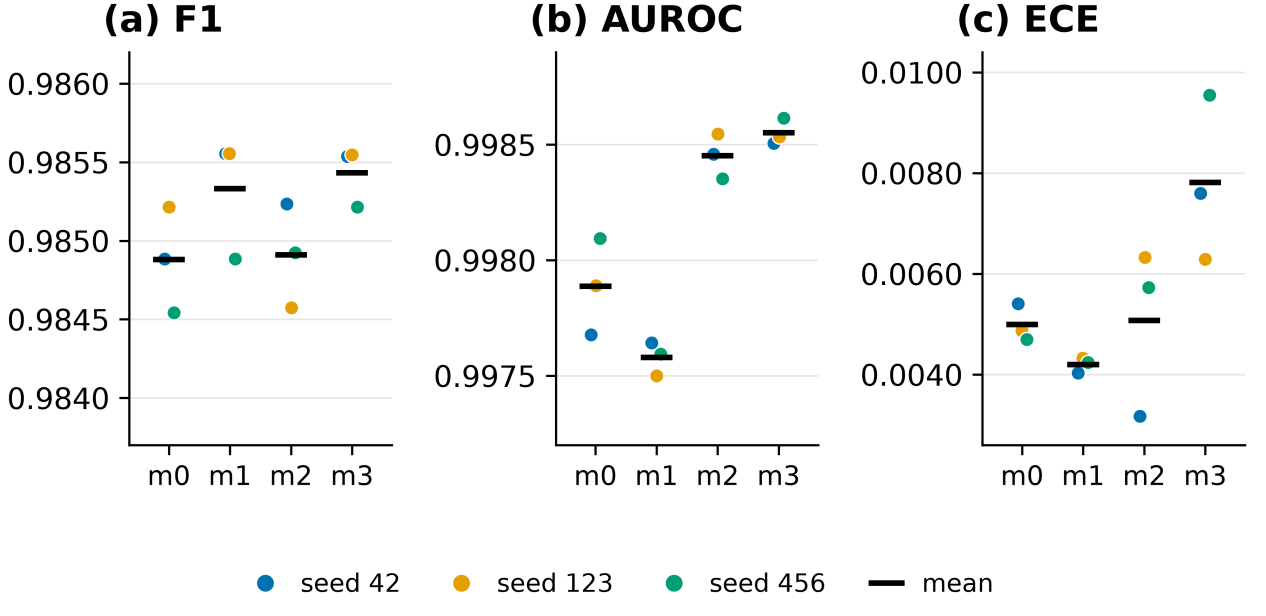


Figure 4: Per-seed F1, AUROC and ECE for the fixed-hyperparameter B6 variants.

5.3. Calibration, Shift, and External Validity

The raw, Platt-scaled, and isotonic versions of the conventional model have ECE values of 0.0045, 0.0091, and 0.0070 on HaluEval, showing that the calibrated versions do not decrease ECE in this sample. Figure 5 clarifies this result. On HaluEval, the points at the two ends lie on the diagonal; the middle range of scores contains few distinct values, leading to only a small number of stable bins. FaithBench is aggregated because its scores concentrate near 1.0. The Platt and isotonic curves overlap the raw curve at both ends. The m3 raw score is already well calibrated in-domain, with ECE 0.0078, providing little room for improvement in calibration on HaluEval. Platt scaling applies a two-parameter logistic transformation mapping the uncalibrated probability s (the raw score in the figures) to a probability,

$$\hat{p} = \sigma(as + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{13}$$

where a is the calibration slope and b is the intercept. A slope below one means the raw score is overconfident, and a slope near one means the scale is already usable. The Brier skill score (BSS) compares the calibrated score with a constant predictor at the evaluation prevalence,

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS}_{\mathrm{prior}}}. \tag{14}$$

A positive BSS means the score beats the prior, and a non-positive BSS means it does not.

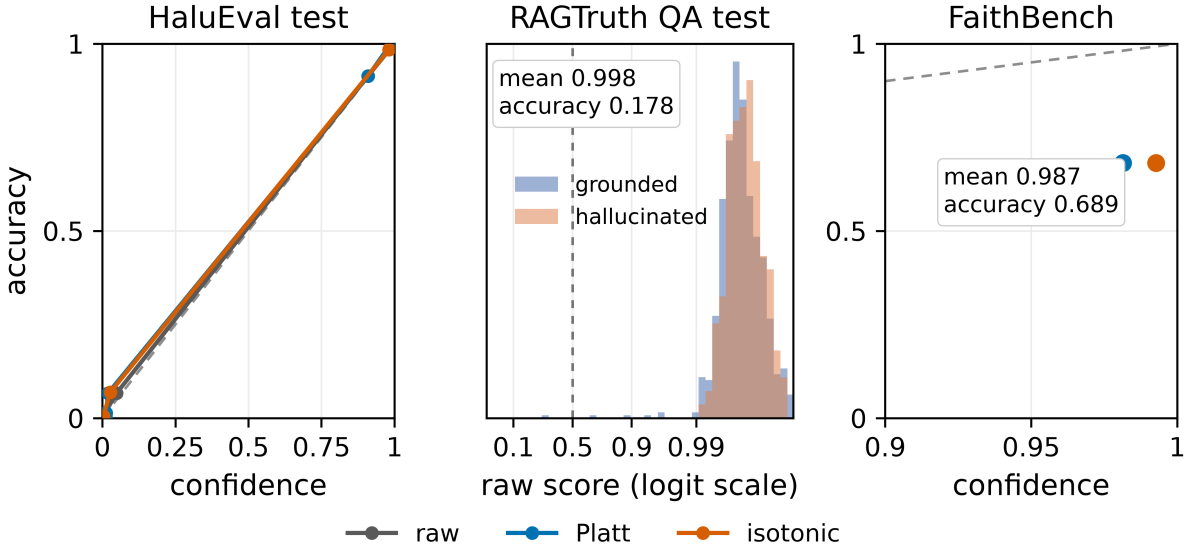


Figure 5: Reliability diagrams for the B2 model on HaluEval and FaithBench, and the RAGTruth QA raw-score distribution.

Source Platt cannot repair the base-rate shift. The fixed-hp m3 raw score is severely overconfident on RAGTruth QA: ECE 0.731 and Brier 0.689, with a Cox slope of 0.18 on the 5,034 calibration rows. Target Platt lowers ECE to 0.125 and Brier to 0.161, but the Brier skill score against the test prior is −0.10, and a constant predictor at the test prevalence has Brier 0.146, lower than every calibrated row. Calibration fixes the scale, not the ranking; AUROC stays at 0.558. The conventional target-Platt row behaves the same way at ECE 0.134 and Brier 0.164. Figure 6 shows the raw and target-recalibrated scores. The source-only operating point (P0) selects the threshold on the HaluEval validation split at a 5% false-positive budget. On HaluEval test, recall is 99.0–99.5% and FPR stays near 5%, but on RAGTruth QA the same threshold flags all 900 rows of the deployed seed-42 checkpoint, because every score exceeds 0.095, so P0 is not usable. Table 8 reports both models on the same test set. A prevalence-only predictor at 0.1778 reaches ECE 0.000 and Brier 0.146 and predicts no positives; at the calibration prevalence of 0.3107 it gives ECE 0.133 and Brier 0.164, so the calibrator sets the level to the calibration prevalence and the residual ECE is dominated by the prevalence gap. The conventional target-thresholded mode (P2) selects 0.313 at a 5% false-positive budget and reaches 5.0% recall

at 5.8% FPR (precision 15.7%, F1 0.076). At 17.8% prevalence, random flagging at 5.8% FPR yields 5.8% recall, so P2 is chance-level. At 0.5, P1 flags no RAGTruth QA rows; for the deployed seed-42 calibrator, the maximum target-Platt score is 0.355 (35.4% before rounding, 35% on the card), the median is 0.331, and 16% of QA rows reach 0.35, so the card usually relies on the per-claim verdict adjustment. Figure 7 shows the operating point and label budget: mean ECE is 0.130 at 50 rows, 0.124 at 252 rows, and 0.125 at 5,034 rows, but a single 252-row draw can miss the full-split ECE by about 0.06. All calibration metrics refer to the pre-verdict score; the displayed percentage is a design aid, not a validated probability.

Table 8
Calibration of fixed-hp m3 and conventional target-Platt on the RAGTruth QA test set (n = 900, 17.8% positive). *Slope: three-seed mean, bracket: envelope of per-seed 95% intervals.

Predictor	Fit	ECE	Brier	BSS	Slope	Flagged
Prior (test prevalence)	constant 0.1778	0.000	0.146	0.000	—	0.0%
Prior (cal prevalence)	constant 0.3107	0.133	0.164	−0.12	—	0.0%
m3 raw	none	0.731	0.689	−3.71	0.18 [0.12, 0.27]*	96.8%
m3 source Platt	HaluEval val	0.765	0.742	−4.07	—	96.8%
m3 target Platt	RAGTruth cal	0.125	0.161	−0.10	—	0.0%
Conventional target Platt	RAGTruth cal	0.134	0.164	−0.12	—	0.0%

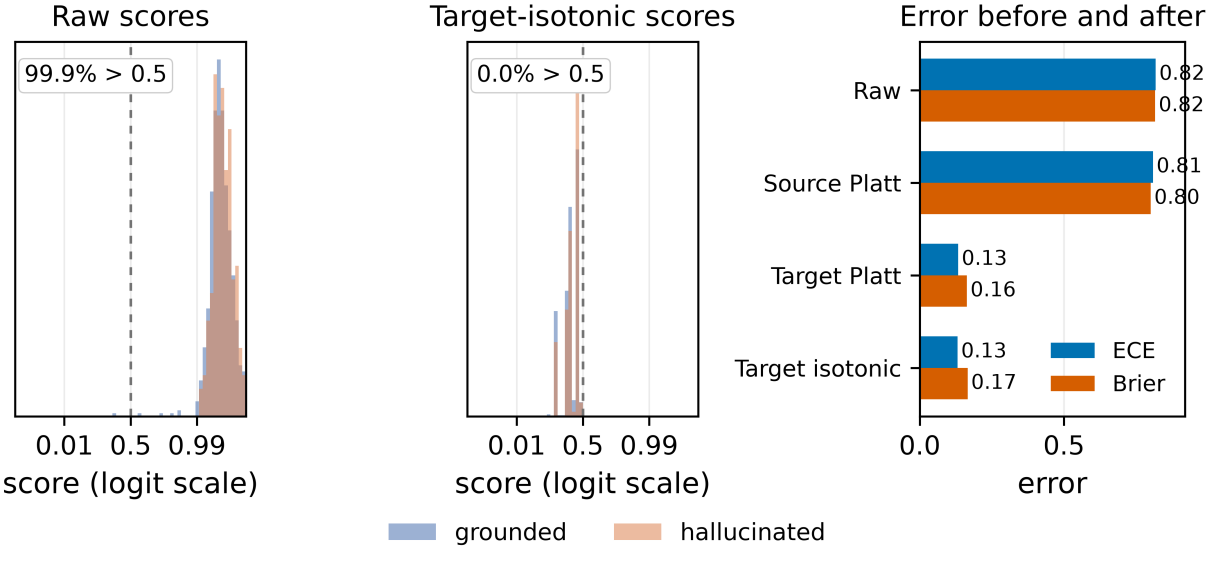


Figure 6: Target-domain recalibration of the conventional model on RAGTruth QA.

Table 9 reports the coverage. Split conformal prediction adds a coverage guarantee to the score. The nonconformity score for label y is $s(x, y) = 1 - \hat{p}_y(x)$, where $\hat{p}_1 = \hat{p}$ and $\hat{p}_0 = 1 - \hat{p}$. The calibration quantile and the prediction set are

$$\hat{q} = s_{(k)}, \quad k = \lceil (n+1)(1-\alpha) \rceil, \qquad C(x) = \{y : s(x, y) \le \hat{q}\}, \tag{15}$$

where $s_{(1)} \le \cdots \le s_{(n)}$ are the ordered calibration scores; this finite-sample order statistic yields coverage of at least $1 - \alpha$ for the prediction set. A class-conditional version replaces the single quantile with one quantile per label,

$$\hat{q}_y = s_{y,(k_y)}, \quad k_y = \left\lceil (n_y + 1)(1-\alpha) \right\rceil, \qquad C(x) = \{y : s(x, y) \le \hat{q}_y\}, \tag{16}$$

where $s_{y,(1)} \le \cdots \le s_{y,(n_y)}$ are the ordered scores in class y and n_y is the number of calibration rows with label y. The main run fits the Platt map and the quantile on the same 5,034 rows, so its coverage is a construction check; it

gives 0.940 and 0.967 coverage on QA and 0.900 and 0.950 on the calibration rows. A disjoint sensitivity fits Platt on 2,514 rows and the quantile on 2,520 rows. On that split, the 900 QA test rows reach 0.944 coverage at 90%, so the marginal guarantee holds. The marginal guarantee hides the class split: positive coverage is 0.685, while negative coverage is 1.000. A class-conditional quantile on the full calibration rows raises positive coverage to 0.887, still slightly below nominal. FaithBench coverage falls to 0.320 and positive coverage to 0.003, because a Platt map fitted on QA sends almost every FaithBench score to the negative side. Conformal prediction therefore certifies the marginal rate only when the calibrator and the quantile are fitted on disjoint rows, and it does not transfer across the task shift. Multicalibration by answer length and claim count shows where the average ECE hides error. On the 900 QA test rows, the target-Platt score has overall ECE 0.125. The worst length bin is 60 words or fewer (n=154), with ECE 0.275; the worst claim-count bin is a single claim (n=36), with ECE 0.346. The Brier score does not follow the ECE ranking, so the average is not a safe summary for short or one-claim answers. FaithBench is not improved by the QA calibrator: its target-Platt ECE is 0.629 against 0.453 for the raw score.

Table 9
Conformal coverage on RAGTruth QA and FaithBench, mean over seeds 42/123/456. Class-conditional rows use the full 5,034 rows; split rows use the disjoint halves.

Dataset	Method	α	Coverage	Pos.	Neg.	Set size
QA test	split	0.10	0.944	0.685	1.000	1.59
QA test	split	0.05	0.972	0.842	1.000	1.75
QA test	class-cond.	0.10	0.886	0.887	0.886	1.69
QA test	class-cond.	0.05	0.936	0.925	0.938	1.80
FaithBench	split	0.10	0.320	0.003	1.000	1.00
FaithBench	split	0.05	0.323	0.006	1.000	1.01

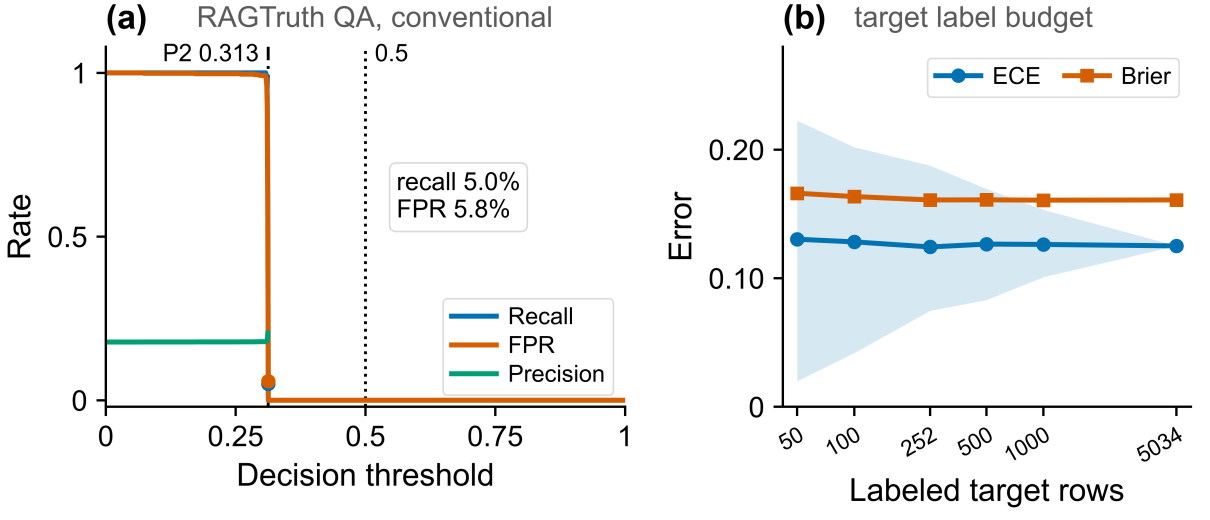


Figure 7: Operating point for the conventional target-Platt model and 100-draw label budget for the fixed-hp m3 calibrator on RAGTruth QA.

The conventional reference m0 is trained only on HaluEval, whereas EC-XGB also sees RAGTruth summarization and data-to-text rows, so those two pairs measure adaptation, not general shift (Table 10). On the pooled RAGTruth test split, the flag rate falls from 99.9% to 61.3% and AUROC rises from 0.475 to 0.582. The gain is task-specific: summarization AUROC rises from 0.634 to 0.736 and data-to-text from 0.567 to 0.771. Held-out QA remains near chance (AUROC 0.547 to 0.549). FaithBench AUROC rises from 0.509 to 0.594, but F1 falls from 0.813 to 0.120 and raw ECE rises from 0.292 to 0.450, so the raw scores transfer poorly. The conventional F1 matches the all-positive

baseline because it flags almost every row. ECE uses raw scores, F1 uses a fixed 0.5 threshold, and Flagged is the share above it. Tables 7, 8, 11, and 17 report the fixed-hp refit; Table 10 reports the tuned B6 checkpoint.

Table 10
External and adaptation performance of B6 m0 and EC-XGB m3 from the tuned B6 run, mean over three seeds.

Dataset / task	Model	F1	AUROC	PR-AUC	Flagged	ECE
RAGTruth pooled	Conventional (m0)	0.5181	0.4753	0.3139	99.9%	0.6346
RAGTruth pooled	EC-XGB (m3)	0.5608	0.5815	0.3588	61.3%	0.2782
RAGTruth QA	Conventional (m0)	0.3022	0.5465	0.1911	99.9%	0.8149
RAGTruth QA	EC-XGB (m3)	0.3081	0.5491	0.1884	97.4%	0.7433
RAGTruth summarization	Conventional (m0)	0.3702	0.6343	0.3536	99.8%	0.7499
RAGTruth summarization	EC-XGB (m3)	0.3434	0.7358	0.4449	11.6%	0.0519
RAGTruth data-to-text	Conventional (m0)	0.7830	0.5665	0.6831	100.0%	0.3390
RAGTruth data-to-text	EC-XGB (m3)	0.8234	0.7705	0.8374	74.9%	0.0527
FaithBench	Conventional (m0)	0.8130	0.5089	0.6735	99.5%	0.2917
FaithBench	EC-XGB (m3)	0.1201	0.5941	0.7109	7.3%	0.4500

The task-shift test is the weakest. On dialogue, m3 flags 99.3% of rows and its F1 equals the all-positive baseline, with AUROC 0.763; the length-only baseline reaches 0.724. On summarization, m3 reaches AUROC 0.717 and flags 37.5%, while the length-only baseline reaches 0.788 and flags every row. The conventional m0 model flags 99.2% of dialogue rows and 97.7% of summarization rows, with AUROC 0.741 and 0.605. On dialogue, both models collapse to the all-positive baseline in F1, though their AUROCs stay above chance; on summarization, both rank below the length-only model. Figure 8 shows the raw score distributions of the conventional model: the HaluEval panel separates the classes, while the external panels pile both classes near 1.0, so the 0.5 threshold flags almost every row. This is the visual form of the 99.9% flag rate in Table 10.

Table 11
External validity on corpora never used for training, calibration, threshold selection, or model choice.

Dataset (n)	Model	F1	AUROC	ECE	Flagged	Pair acc.
MedHallu (2,000)	EC-XGB (m3)	0.667	0.821	0.493	100.0%	0.867
	Conventional (m0)	0.667	0.781	0.499	100.0%	0.822
	Length-only	0.667	0.260	0.500	100.0%	0.200
HaluBench (900)	EC-XGB (m3)	0.588	0.600	0.334	50.3%	—
	Conventional (m0)	0.585	0.609	0.346	51.6%	—
	Length-only	0.441	0.514	0.420	33.2%	—
HaluEval dialogue (2,000)	EC-XGB (m3)	0.670	0.763	0.488	99.3%	—
	Length-only	0.680	0.724	0.438	95.9%	—
HaluEval summarization (2,000)	EC-XGB (m3)	0.583	0.717	0.113	37.5%	—
	Length-only	0.667	0.788	0.500	100.0%	—

Table 11 reports the two external corpora and the task-shift pair. MedHallu is the strongest evidence that the ranking survives a domain change and the clearest evidence that the probability does not. The fixed-hp m3 model ranks the paired hallucinated answer above the grounded answer in 86.7% of 1,000 medical pairs [84.5, 88.7], with AUROC 0.821 [0.806, 0.837] on the 2,000 rows. The pair accuracy over M pairs is

$$\mathrm{PA} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\left[\hat{p}(a_m^+) > \hat{p}(a_m^-)\right], \tag{17}$$

where a_m^+ is the hallucinated answer and a_m^- the grounded answer. The conventional m0 model reaches 82.2% and 0.781. The length-only baseline is inverted, at 20.0% and 0.260, because grounded MedHallu answers are longer than

the paired hallucinations (39.7 vs 27.8 words), the opposite of HaluEval (2.2 vs 11.0). The ranking therefore transfers, but the scale does not: m3 flags every row, so its F1 is the all-positive 0.667 and its ECE is 0.493. The source indicator is one for MedHallu and zero for HaluBench; a source-zero sensitivity on MedHallu changes pair accuracy by at most 0.003 and AUROC by at most 0.002, so the indicator does not drive the result. HaluBench is a 30% stratified sample. Its overall AUROC is 0.600, with covidQA 0.833, DROP 0.513, and FinanceBench 0.527; the model transfers to the biomedical subset but is at chance on the other two. The length-only baseline is at 0.514 overall, so 0.600 is not a length artifact. All evaluation sets are balanced 50/50 by construction, so F1 and the flag rate are prevalence-dependent and do not describe natural deployment; HaluBench is unpaired, so pair accuracy is undefined, and the length-only baseline is fit once on HaluEval train only.

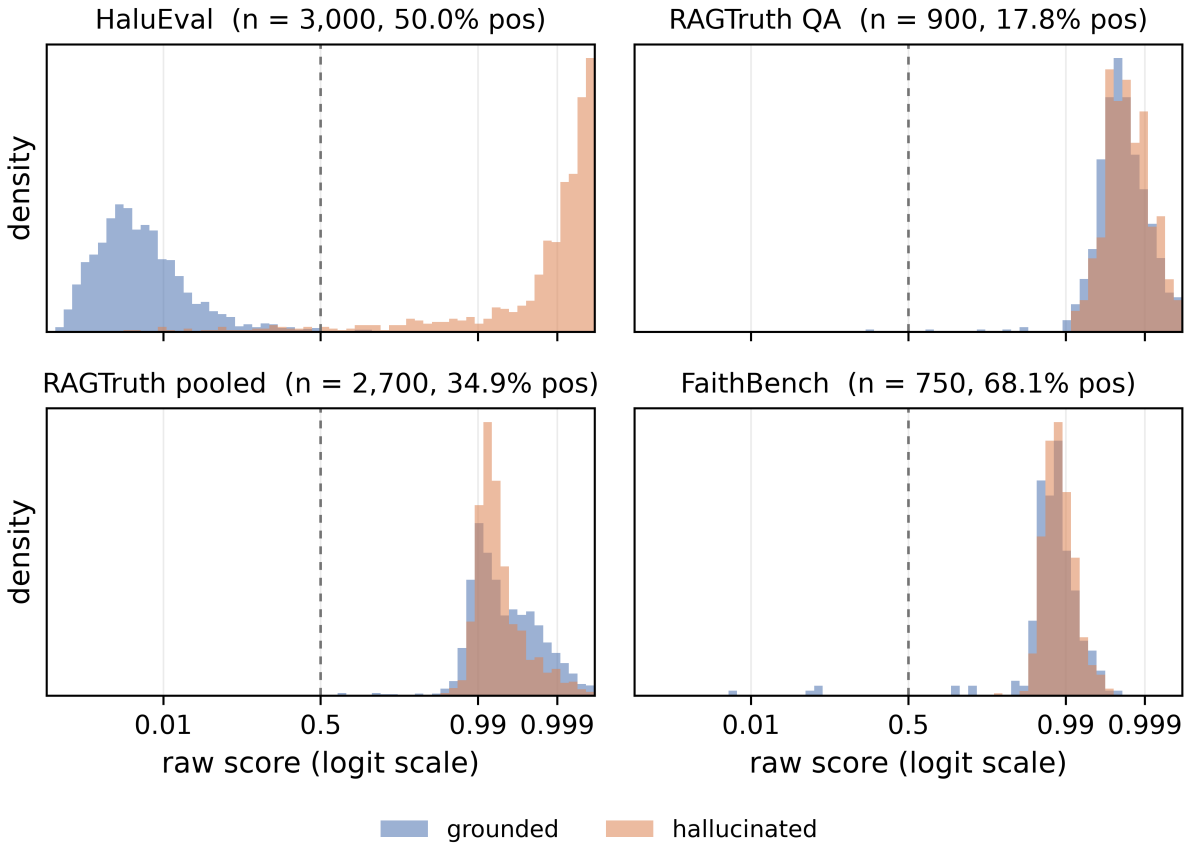


Figure 8: Raw score distributions of the conventional B2 model on HaluEval and the external test rows.

An index stress test asks whether the weak held-out QA result is a retrieval problem. The hybrid BM25 and FAISS index covers the 989 RAGTruth QA source documents. On a 300-row, 50-group sample, recall@1 is 0.94, recall@5 and recall@10 are 1.00, and MRR is 0.959. Replacing the gold context with the top-5 retrieved passages changes AUROC from 0.545 to 0.521, F1 from 0.345 to 0.349, and ECE from 0.706 to 0.704. The AUROC delta is −0.023 [−0.061, 0.011], which includes zero. Retrieval is therefore not the bottleneck: the model is near chance on RAGTruth QA even with the gold source passage. The qrels are the known source passage, not human web-search judgments, so this measures index recall, not end-to-end search quality. The sample has 50 source groups, so the interval is wide.

5.4. Statistical Analysis

Statistical tests use the paired fixed-hyperparameter predictions and 2,000 two-level bootstrap draws, where groups are resampled within each seed and then seeds are resampled. McNemar's test compares the discordant counts b and c on the paired seed-42 rows, where b is the number of rows where the first model is right and the second is wrong, and c is the reverse; the statistic uses a continuity correction. Holm's method controls the family-wise error rate across

the four baseline comparisons. The 95% interval is the 2.5th and 97.5th percentile of the paired bootstrap deltas. The two-level version resamples groups within each seed before resampling seeds. The Wilcoxon signed-rank test compares the two models across the three seeds. McNemar's results are presented in Table 12. Neither EC-XGB nor the conventional method exhibits a significant difference under the deployed checkpoints ($p = 0.45$), whereas the fixed-hp ablation shows $p = 0.789$ on the same test rows. The fixed-hp m0-vs-m3 comparison is the primary test, and it is not part of the Holm family. The family consists of four baseline comparisons against the conventional method. The test uses the single configuration defined in Table 6. The random forest shows no significant difference from the conventional method ($p = 0.502$). Logistic regression and the heuristic significantly underperform ($p = 1.86 \times 10^{-10}$ and $p = 2.28 \times 10^{-30}$), with both surviving the Holm correction. The Wilcoxon test with three seeds does not produce a significant difference (minimum p-value = 0.25). Table 13 contains the results of the two-level bootstrap evaluation. On the HaluEval dataset, the F1 delta for m3 minus m0 is +0.0006, with CI [−0.0010, 0.0021], which encompasses zero, whereas the AUROC delta is +0.0007, with CI [0.0002, 0.0012], which excludes zero by a small margin. The F1 deltas for m2 minus m1 and m3 minus m1 are −0.0004 (CI [−0.0020, 0.0012]) and +0.0001 (CI [−0.0015, 0.0018]). Thus, the claim features provide no overall improvement in terms of F1. The AUROC deltas are +0.0009 and +0.0010, with both intervals above zero.

In the held-out RAGTruth QA experiment, the m3−m0 F1 difference is +0.0076 [0.0045, 0.0109], while the AUROC difference is +0.0017 [−0.0270, 0.0291], which includes zero. The F1 gain is achieved mainly by flagging almost every row, not by improving the ranking. In the pooled seen tasks experiment, the F1 difference is +0.0451 [0.0348, 0.0554], and the AUROC difference is +0.0962 [0.0644, 0.1295]. All the paired fixed-minus-tuned differences include zero, meaning that the rebuild does not affect these results. In HaluEval, the groups are item pairs (1,500); in RAGTruth QA, there are 150 source groups; and in the pooled split, there are 450 groups. The QA AUROC difference includes zero, suggesting that the held-out transfer result is unresolved, and the pooled row measures adaptation to seen tasks.

Table 12
McNemar tests on the paired seed-42 HaluEval predictions.

Comparison	b	c	p	Holm p
EC-XGB vs conventional	10	6	0.453	—
EC-XGB vs conventional (fixed hp)	8	6	0.789	—
Conventional vs random forest	12	8	0.502	0.502
Conventional vs logistic regression	69	11	1.86×10^{-10}	3.7×10^{-10}
Conventional vs heuristic	158	9	2.28×10^{-30}	6.8×10^{-30}
Conventional vs NLI-only	1006	26	5.6×10^{-204}	2.2×10^{-203}

Table 13
Two-level bootstrap deltas for the fixed-hyperparameter ablation (2,000 draws; groups within seed, then seeds). Intervals are 95%.

Dataset	Comparison	ΔF1 [95% CI]	ΔAUROC [95% CI]
HaluEval	m3−m0	+0.0006 [−0.0010, 0.0021]	+0.0007 [0.0002, 0.0012]
HaluEval	m2−m1	−0.0004 [−0.0020, 0.0012]	+0.0009 [0.0004, 0.0014]
HaluEval	m3−m1	+0.0001 [−0.0015, 0.0018]	+0.0010 [0.0004, 0.0016]
RAGTruth QA	m3−m0	+0.0076 [0.0045, 0.0109]	+0.0017 [−0.0270, 0.0291]
RAGTruth pooled	m3−m0	+0.0451 [0.0348, 0.0554]	+0.0962 [0.0644, 0.1295]

5.5. Explanation, Efficiency, and Judges

The explanation uses the fixed-hyperparameter m3 seed-42 checkpoint. Figure 9 lists the top-15 features ranked by mean absolute SHAP value over 1,000 stratified HaluEval test rows. Context overlap is the strongest driver, followed by word count, question-answer cosine, character count, and question-answer Jaccard similarity. The claim-contradiction mean appears at rank eight. The top ten features cover 77.9% of the attributions, and exact TreeSHAP matches the model margin up to 9.2×10^{-6}. The attributions are calculated for the raw margin, not the display score. The m3

rankings are highly correlated with the frozen B2 rankings (Spearman's $\rho = 0.878$, Kendall's $\tau = 0.730$); therefore, the claim features refine the explanation, rather than replace the original drivers. Cross-seed Spearman correlation is 0.972 and 0.950; nine of the top ten features are shared, and the top five features are identical. The explanation is resilient to the choice of the seed but not to content editing. Table 14 presents a 20-case perturbation pilot: content edits move the score, structural edits barely shift it, and the top-1 feature is flipped in 54.5% of entity perturbations and none of the other operators. The rate of entity flipping is higher than in B2, so the explanation is less stable when the answer entity changes; the per-operator sample size is small. The structured AI-assisted audit of 26 cases enriched for errors and re-scored under m3 evaluated each explanation based on consistency with answer-context evidence, rather than the dataset label. Thirteen explanations were considered implausible, three uncertain, and ten plausible, so the explanations are used as diagnostic aids, not faithful attributions. False positives carry the length confound, and false negatives frequently rely on entities repeated in the context. As the sample is enriched for errors, the percentages characterize the audit, not the test set. The labels are author-checked pre-annotations: LLM tools assisted the annotation, and the authors reviewed and verified the final labels. Agreement with the pre-annotations was 0.98 for explanation, 0.94 for taxonomy, and 0.88 for claims. Cohen's κ was 0.965, 0.923, and 0.788; the authors changed 2%, 6%, and 12% of rows, and the blind Fleiss κ was 0.940, 0.868, and 0.761.

Table 14
Perturbation stability of the m3 raw score and SHAP ranking on HaluEval (seed 42, 20-case pilot).

Operator	n	Δscore	Large rate	Top-1 flip	Spearman
Entity replacement	11	0.339	0.364	0.545	0.848
Support removal	10	0.464	0.500	0.000	0.808
Number replacement	3	0.332	0.333	0.000	0.731
Date replacement	2	0.482	0.500	0.000	0.786
Irrelevant insert	20	0.003	0.000	0.000	0.922
Clause shuffle	2	0.000	0.000	0.000	0.999

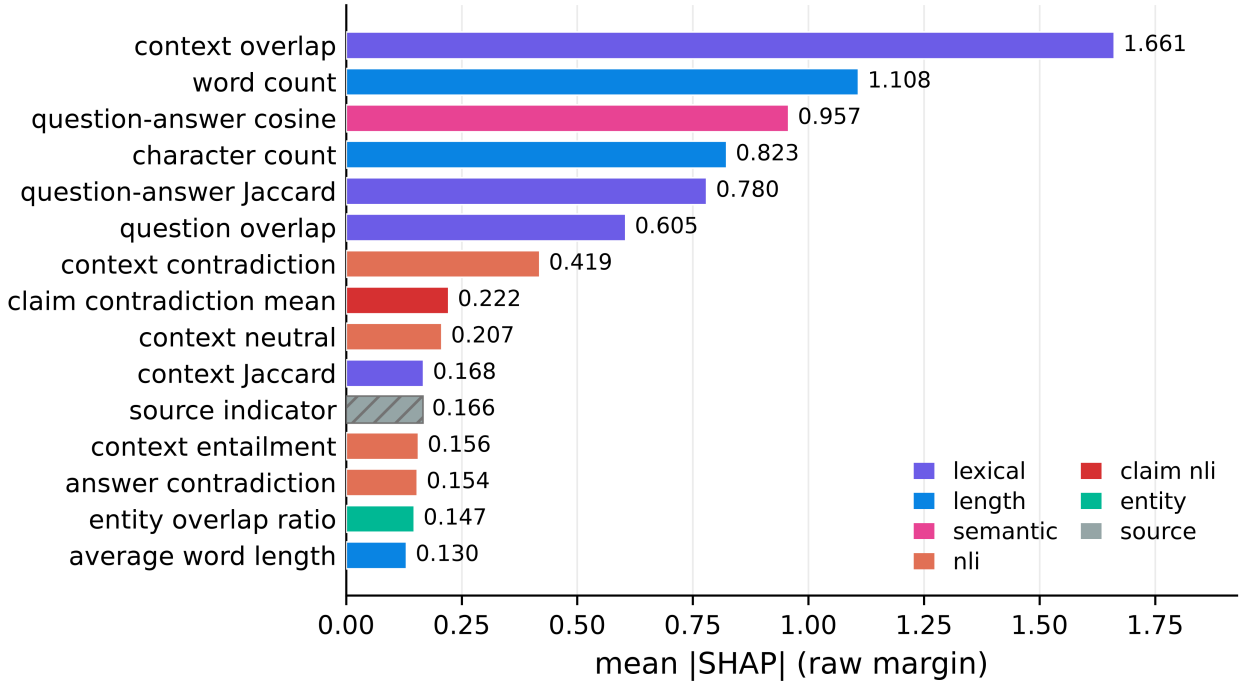


Figure 9: Mean absolute TreeSHAP for the fixed-hp m3 model (top 15 features).

Table 15 contains component latencies. Components are measured separately in instrumented passes over the same samples and hence do not add up to the end-to-end median, and retrieval and judge calls are excluded. NLI and NER dominate the base median, while XGBoost and SHAP contributions are negligible. Including the eight claim features increases the m3 pipeline median to 127.5 ms. The same 200-row protocol takes 2.00 s per answer in a single contended run (p50 0.98 s, p95 7.56 s) on CPU; therefore, this value depends on GPU feature extraction, which is the bottleneck here, not the model. Table 16 shows the comparison of cost and latency: classical heads run in less than a millisecond, while the two judges are 18 to 24 times slower than the m3 pipeline. The encoder takes 6.97 ms per row at batch size 8 and fp16 precision, making the feature pipeline approximately 18 times slower, but the encoder requires a fine-tuned classification head, and its cost is not included. Model-only m3 inference is 0.008 ms, and the NLI-only head is the NLI step and a linear head. Size denotes the size of the seed-42 artifact. Cost is calculated with token counts and an assumed price of $1.20 per million output tokens, and the encoder time is at 512 tokens as an upper bound for the 256-token baseline.

Table 15
Per-component latency in milliseconds (n = 200).

Component	p50	p95	mean
Core lexical	0.10	0.15	0.11
Entity (spaCy NER)	21.13	47.24	25.34
NLI (2 directions)	27.09	68.60	40.09
Semantic (MiniLM)	9.25	15.55	12.56
XGBoost inference	1.26	8.80	2.62
SHAP explanation	1.91	15.52	3.65
Base total	61.77	—	—
Claim NLI (m3)	43.75	83.21	53.11
Total m3	127.51	234.59	145.89

Table 16
Per-baseline cost and latency.

Baseline	Inputs	Size	p50 (ms)	Cost/1K
Heuristic (overlap)	text	—	0.009	≈$0
Logistic regression	26 feats	1.1 KB	0.001	≈$0.001
Random forest	26 feats	8.4 MB	0.331	≈$0.001
NLI-only	6 NLI probs	—	—	≈$0.001
TF-IDF (answer/full)	text	0.58/4.1 MB	0.014/0.110*	≈$0.001
XGBoost (conventional)	26 feats	0.49 MB	0.005	≈$0.001
RoBERTa-base	512 tokens (timing)	125M params	6.97	training only
EC-XGB (m3)	35 feats	0.68 MB	127.51	≈$0.001
GPT 5.6 Luna judge	API	—	2283.07	$0.234
Qwen 3.5 4B judge	4-bit local	4B params	3049.96	$0

LLM judges are not fair evaluators, and their preferences depend on the answer order [38]. Table 17 compares EC-XGB with two judges on the RAGTruth QA rows at natural prevalence. GPT-5.6 Luna has AUROC 0.910 compared to 0.558 for EC-XGB, and Qwen 3.5 4B also beats EC-XGB, meaning that both judges are superior QA detectors, while the feature model advantage is only in cost. The target-fitted length-only baseline also ranks the QA rows higher than the feature model. Under matched multi-source training, the adapted encoder achieves 0.694 on the pooled RAGTruth test set, outperforming EC-XGB at 0.582. The feature model has less degradation on new tasks, but both models approach random guessing. The length-matched panel changes the ranking: on the exact-length subset, EC-XGB performs similarly to the stronger judge in AUROC and better in F1, meaning that the natural-prevalence difference is due to the domain shift and not the length difference. The paired interval for GPT minus EC-XGB is −0.052 [−0.127, 0.024], and the AUROC interval is +0.008 [−0.034, 0.048]. Both include zero. Prompt choice moves GPT's F1 from

0.814 to 0.854 across three prompts, about the size of the model gap. GPT uses default decoding and three seeds, while Qwen is a single deterministic run, so the judge ranking is not stable.

Table 17
Judge stress tests on the 900 RAGTruth QA rows and the 158-row length-matched HaluEval set.

Model	P	R	F1	AUROC	Flagged	p50 (ms)	Cost/1K
GPT 5.6 Luna judge	0.404	0.942	0.565	0.910	41.4%	2283	$0.234
Qwen 3.5 4B judge	0.570	0.638	0.602	0.778	19.9%	3050	$0
EC-XGB (m3)	0.183	0.998	0.310	0.558	96.8%	127.5	≈$0.001
Length-only	0.439	0.181	0.257	0.712	7.3%	—	≈$0
All-positive	0.178	1.000	0.302	—	100.0%	—	≈$0
Length-matched HaluEval, 158 rows, 50% positive							
GPT 5.6 Luna judge	0.945	0.798	0.865	0.957	42.2%	2283	$0.234
Qwen 3.5 4B judge	1.000	0.696	0.821	0.863	34.8%	3050	$0
EC-XGB (m3)	0.954	0.865	0.907	0.956	45.4%	127.5	≈$0.001
Length-only	0.515	0.443	0.476	0.521	43.0%	—	≈$0

5.6. Error Analysis and Discussion

The m3 seed-42 model makes 43 errors in HaluEval (Figure 10), which amounts to an error rate of 1.4%. WP5 includes seven out of eight false positives and 12 out of 35 false negatives chosen to cover the codebook. An extra 50-instance worksheet, reviewed by one author, brings 31 artifact cases and 19 content cases distributed into 12 categories. These records are about the instances being reviewed, not the entire test set of 3,000 rows. Table 18 shows that, for the conventional B2 model, length is essentially a perfect indicator on the HaluEval benchmark: false positives tend to occur in the five-to-eight length bin and false negatives tend to occur in the shorter bins, although there are only two negative rows in the 17+ bin. Both controls affect the length distribution. First, the reweighting control ensures equal weighting across six length bins, each containing 247 instances. Second, the length-matching control matches one positive instance and one negative per each length value. Reweighting results in an improvement of 0.039 for F1 and 0.023 for AUROC in EC-XGB in contrast with length-only regression, which is already competitive on its own (Table 5). Length matching gives 79 pairs (158 instances) over 1,000 bootstraps. EC-XGB holds AUROC of 0.956 whereas the length-only baseline falls to 0.521. The gap in AUROC between these two models is 0.435 (the bootstrap mean is 0.436 with 95% confidence interval of [0.351, 0.522]), and the gap in F1 is 0.431 (the bootstrap mean is 0.440 with 95% confidence interval of [0.334, 0.552]).

Table 18
HaluEval test errors of the conventional B2 model by answer length (counts seed 42, rates three-seed means).

Words	Rows	Hallucinated rate	FP rate	FN rate	Mean score
1	524	0.021	0.004	0.182	0.024
2	514	0.033	0.000	0.176	0.035
3–4	474	0.127	0.002	0.178	0.126
5–8	641	0.902	0.106	0.031	0.888
9–16	600	0.982	0.000	0.003	0.980
17+	247	0.992	0.333	0.000	0.993

Figure 11 demonstrates the same pattern. The long complete sentences are misclassified by the conventional model as being hallucinated, while the in-domain model works mostly as a length and overlap detector. In published HaluEval QA results different splits, supervision, and evidence access are used. A supervised 13B agent achieves F1 of 0.724 on 100 instances, so these numbers are context and cannot be used as a head-to-head comparison. The benchmark also constructs hallucinated answers about five words longer, while the observed gap is 8.5 words, so any comparison that ignores this construction artifact overstates semantic detection. The claims of the current work are bounded by the experiments that have not been performed. The retrieval quality and end-to-end verdict accuracy have not been

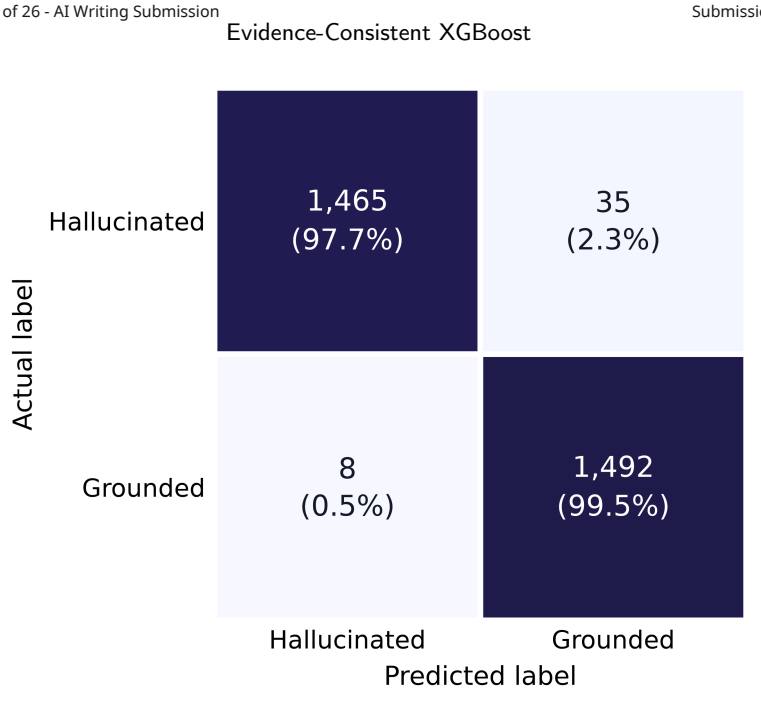


Figure 10: Confusion matrix for the seed-42 fixed-hp m3 model on HaluEval at the 0.5 threshold (n = 3,000). Cells show counts with row-normalized percentages.

estimated, the judge comparison controls length but not domain, and the audit is error-enriched. The deployed display calibrator was selected in an earlier fit that compared Platt and isotonic on the test split, and Table 8 refits Platt without that selection.

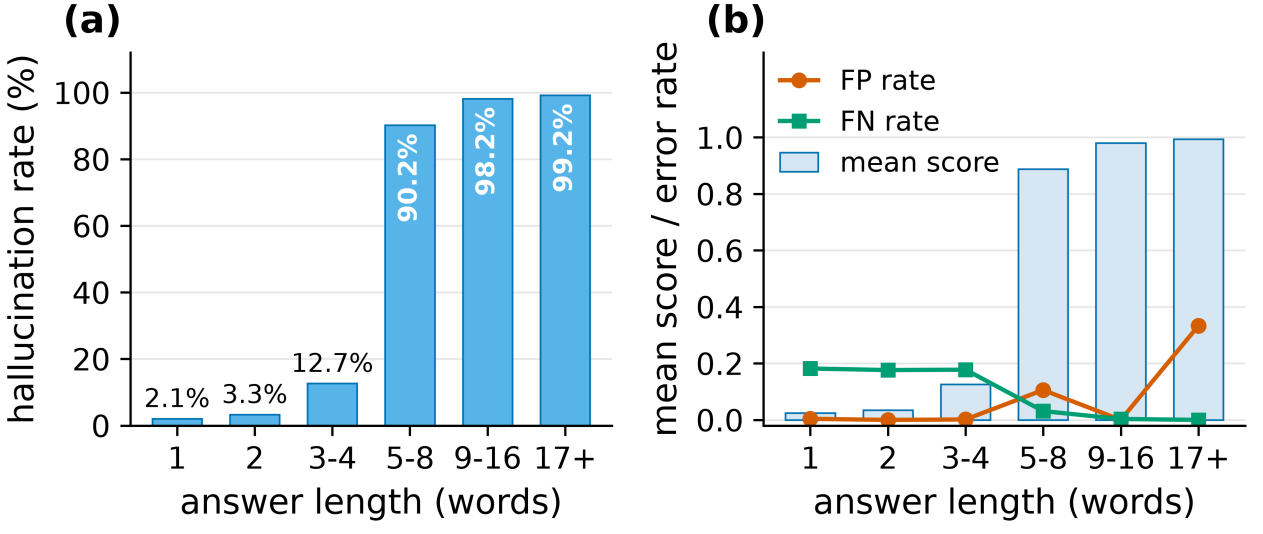


Figure 11: Hallucination rate, mean score, and error rates by answer length for the B2 conventional model.

6. Conclusion

The current paper tests EC-XGB, an evidence-consistent, 35-feature model, as a hallucination risk score and how its calibration and ranking change under domain and task shifts. The model's ranking is still intact after the corpus has changed and allows the score to rank answers for review. Moreover, its per-claim verdicts provide a reviewer with the explanation of each flagged instance. On the other hand, its probability is not so resilient; the display score requires

labeled target rows, and target prevalence dictates the score's value. As a result, EC-XGB operates as a domain-specific screening score, not a general-purpose detector. The upcoming assessment considers adaptation to an unseen question-answering domain under prevalence shift. The seen-task source ablation will separate RAGTruth rows from the source indicator.

CRediT authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Saiful Alam Sabbir:** Data curation, Validation, Writing - review & editing. **MD. Miraz Ahamed:** Data curation, Validation, Writing - review & editing. **Ohidujjaman Tuhin:** Supervision, Validation, Writing - review & editing.

Declarations

- **Funding:** This research received no external funding.
- **Conflicts of Interest:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Data availability:** Raw datasets are not redistributed in full. For HaluEval (MIT), RAGTruth (MIT), and FaithBench (CC BY-NC-SA 4.0), the repository provides download instructions and a tracked license manifest of source revisions and SHA-256 hashes, and an audit file with 26 cases consisting of excerpts from the benchmark datasets. MedHallu (MIT) and the HaluBench subset (CC BY-NC 2.0) were obtained from their official releases. The splits and feature matrix can be generated from the original datasets. The model artifacts and the experiment manifest are available from the corresponding author on reasonable request.
- **Code availability:** The training pipeline, FastAPI inference service, and Next.js dashboard are available at `https://github.com/Atik203/HaluRISC` (branch `main`). The code is released under the MIT license.
- **Ethical approval:** Human-subject data were not collected. The authors checked LLM-assisted pre-annotations of public benchmark rows for internal verification; this procedure is not considered human-subjects research, and therefore no ethics approval was necessary.

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